Machine Learning Based Automatic Diagnosis in Mobile Communication Networks

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***Abstract*—The self-healing function in self-organizing networks can not only detect the presence of fault conditions but also diagnose the root causes in a fully autonomous fashion. In this paper, we propose a machine learning based diagnosis algorithm that uses network condition indicators such as key performance indicators and performance management counters for network condition diagnosis. The proposed algorithm judiciously combines the classical supervised softmax neural network (SNN) and support vector machine (SVM), and therefore can be efficiently implemented by off-the-shelf tools while achieving promising diagnosis performance. In particular, the proposed algorithm combines the features extracted from SNN and SVM, and is able to robustly diagnose fault conditions with different levels of severity. Besides, the proposed algorithm can also handle complex scenarios where there is more than one fault condition present at the same time. Considering that data with labels of multiple faults are not available in general, we further propose a simple retraining procedure which allows the proposed algorithm to perform multi-fault diagnosis even when the training data are only single labeled. Simulation results demonstrate that the proposed algorithms provide desired diagnosis performance in both single-fault and multi-fault scenarios and outperform the traditional scoring based methods.**

***Index Terms*—Fault diagnosis, Self-healing, Machine learning, Long-Term Evolution (LTE).**

I. Introduction

WITH the rapidly-increasing complexity of commercial wireless mobile networks and the number of mobile users, it has become a significant challenge for telecommunication operators and vendors to provide high quality of service (QoS) while maintaining the infrastructure of wireless networks. To help reduce both capital investment and operational expenditures, the 3rd Generation Partnership Project (3GPP) [1] has introduced a concept of self-organizing networks (SON) in the Long-Term Evolution (LTE) specification. The SON is identified as a key design principle for next generation networks. Research efforts, such as those in [2]–[6], for automatic configuration, optimization, and fault healing, have been made to achieve the goal of SON. The three main functionalities of SONs are briefly described as follows:

- Self-configuration: Automatic configuration of network elements such as base stations (BSs) are required in different tasks including planning, extension, and upgrade of network terminals [4], [5]. It is mainly activated whenever a new BS is deployed in the system.
- Self-optimization: The self-optimization function will be triggered after the system has been configured correctly. Based on the network situation, certain network parameters and resource allocation are recalculated and adjusted periodically in order to optimize or improve the network service performance [2], [4], [6].
- Self-healing: The increasing complexity of cellular networks calls for an automatic process to handle cell outages or service degradation. Based on collected key performance indicators (KPIs) and performance management counters (PMCs), the self-healing function will be activated whenever a fault or failure occurs. As illustrated in Fig. 1, a self-healing system has three main components: fault detection, root cause diagnosis, and corrective actions. Automatic root cause analysis is challenging in general, and currently it is mainly carried out manually by experts [7]. The main objective of a self-healing function is not only to detect the failure events but also to diagnose the failure, to discriminate the reason, and then to implement the compensation mechanisms, leading the network to function properly.

The objective of this paper is to propose an automatic fault diagnostic method that can be part of a self-healing Long-Term Evolution-Advanced (LTE-A) network. There have been considerable research works for automatic fault diagnosis. In [8]–[10], the use of a Bayesian network (BN) was proposed for automatically diagnosing fault causes for wireless networks. However, these systems cannot be easily deployed since expert knowledge have to be deeply involved in parameter tuning and

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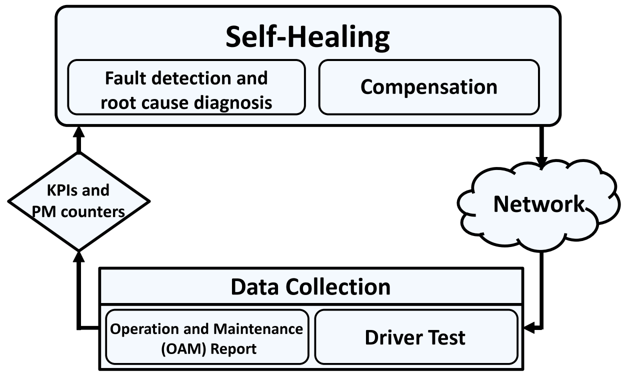


Fig. 1. System diagram of the self-healing function in SON.

threshold setting that are required in the BN based methods. In [11], by assuming that there is a historical database of system KPIs for the normal condition and network faults, the authors proposed the use of a KPI level as a measure of deviation from the normal condition, and the frequencies of KPI occurrence under each fault to calculate a score for each fault. Based on the scores, the system makes a diagnosis decision. The scoring based diagnosis system in [11] thereby involves less domain knowledge.

Aiming at further reducing the involvement of expert knowledge, there has been an increasing interest in applying machine learning (ML) techniques for automatic fault diagnosis, although at the same time ML is also considered in various physical-layer designs [12]–[14]. For example, the works in [15], [16] applied the unsupervised clustering methods for identifying relevant KPIs (i.e., feature extraction) for fault diagnosis. The recent works in [17] and [18] employed self-organizing maps and the stacked denoising autoencoder for dimension reduction, respectively. To be used for learning the diagnosis model, experts are still needed in labeling the clusters and dimension reduced features in corresponding to the fault causes.

In this paper, like [11], we assume that a set of training data about system KPIs and performance management counters are given, and propose an ML based automatic diagnosis algorithm. One of the merits of the proposed diagnosis algorithm lies in the consideration of two practical issues that are often overlooked in the previous works. The first issue is about the robustness of the diagnosis algorithm against scenarios with different levels of severity of fault cases. It is desirable that the diagnosis algorithm is able to detect the fault cause even when the fault is slight and not severe. The second issue is about the fact that in practice there could be multiple faults occurring at the same time, and it is desirable that the diagnosis system can report the multi-fault condition, even when the training data are single labeled only. Another merit of the proposed diagnosis algorithm is that it relies on the classical ML classification techniques (specifically, the feedforward neural network (FNN) and the support vector machine (SVM)), and therefore the training can be implemented efficiently and by off-the-shelf ML tools [19]. The proposed contributions can be summarized as follows.

- Inspired by the fuzzy clustering algorithms [20], we propose a preprocessing technique that automatically weights the relevant KPIs and PMCs for improving the separability between different network conditions. The weights are purely learned from the training data without the need of any expert knowledge. Simulation results will verify that the proposed preprocessing technique can considerably improve the network diagnosis performance.
- To leverage the fact that FNN is a powerful model especially when the feature size is large whereas SVM, as a large margin method, is more robust against non-separated data, we propose a new diagnosis algorithm that judiciously concatenates the softmax neural network (SNN) and SVM. In particular, the SVM and SNN are first used for feature extraction, and the two set of features are properly combined and used for multi-class classification training and testing. As will be explained shortly, such a combined scheme provides a much more robust diagnosis performance and the algorithm is capable of diagnosing slight fault conditions with high accuracy.
- To enable the algorithm to be applied to scenarios with multiple fault conditions occurring at the same time, we replace the regular SVM with the rank-SVM [21] for multi-label multi-class classification. We further consider the scenario where the experts only provide single labels for the multi-fault conditions. To handle this mislabeled case, we propose a simple retraining procedure that uses the output of rank-SVM to correct the labels in the training set and repeat the training once. To the best of our knowledge, this is the first work that attempts to address the multiple fault problem and mislabeled data problem. Simulations results will validate that the proposed diagnosis algorithm can yield desired performance and outperforms the algorithm in [11].

The remainder of this paper is organized as follows. Section II describes the system model and defines the problem formulation for network conditions. The SNN and SVM are also briefly reviewed. In Section III, the preprocessing technique and the two proposed diagnosis algorithms are presented. Section IV presents the empirical diagnosis performance of the proposed algorithms based on a system level simulation of the LTE-A network. Finally, the conclusions are presented in Section V.

II. System Model and Problem Description

We consider the fault detection and diagnosis problem in macro stations based on an LTE-A network [22]. In the system model, we consider only macro stations and ignore base stations of a small cell such as pico, femto, and relay cells because these small cells currently do not have fault detection and diagnosis functions. Fig. 2 presents a layout of an LTE-A homogeneous network with eNodeBs and user equipment (UE). The one-tier interfering cellular network is considered, where each cell is divided into three sectors and each sector is assigned a directional antenna [23]. The transmission modes of eNodeBs are as defined by 3GPP [24]. The transmission modes of all UEs are closed-loop spatial multiplexing. Details of the system parameters and channel models will be carried out in Section IV.

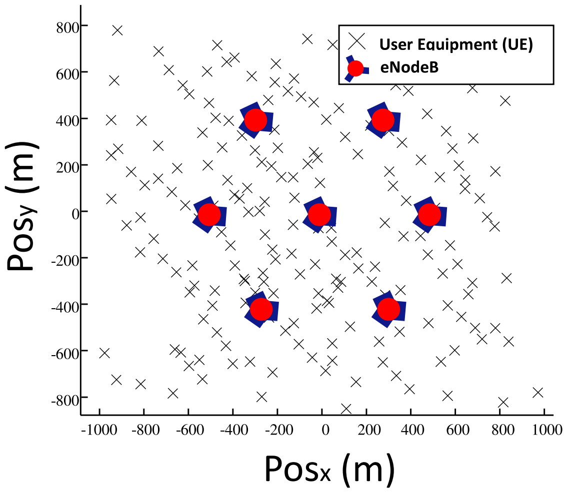


Fig. 2. An LTE-A homogeneous system network.

A. Network Fault Causes

The main objective of the self-healing network is to diagnose fault causes and recover the system to the normal condition. Here, we introduce some of the most relevant fault causes in the cellular networks, namely, excessive uptilt (EU), excessive reduced power (ERP), antenna fault (AF), and coverage hole (CH), in addition to the normal condition. All of them are associated with the quality experienced by the UEs. Each of the network conditions are explained as follows.

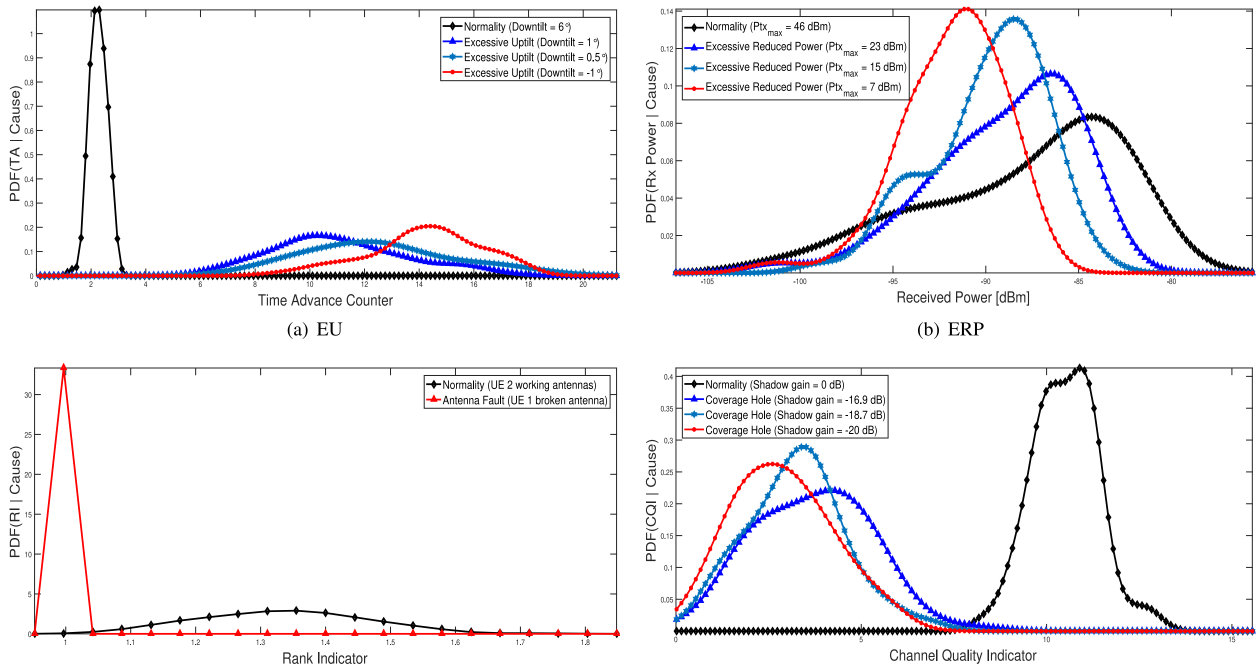
- Normality: Under the normal condition, the network system operates without any fault causes. The mechanical downtilt and the maximal transmission power under normal condition is 6 degrees and 46 dBm, respectively [22]. Moreover, each UE is equipped with two antennas that function normally.
- Excessive uptilt (EU): The EU means the mechanical tilt of the antennas of eNodeBs are increased, and it causes a larger coverage area than under the normal condition.
- Excessive reduced power (ERP): Due to wiring problems or incorrect parameter configurations, the eNodeB can suffer from power supply failure and cause significant reduction of the BS transmission power. The ERP condition results in the receive signal-to-interference-plus-noise ratio (SINR) of UEs in the service area falling below the minimum level. Thus, ERP makes the system fail to maintain the planned performance requirement.
- Antenna fault (AF): Under the normal condition, each UE is configured with two receive antennas. In the case that one of the antennas is broken, the UE is no longer able to support multiple-input multiple-output (MIMO), and would suffer from reduced throughput and data rate.
- Coverage hole (CH): The CH means that there is a region in the service area where the UE does not have enough received power to maintain the service. This CH is caused by geographical effects, that is, physical obstacles such as hills and buildings, that incur increased signal attenuation.

It is worthwhile to mention that, in practice, each of the fault conditions may have different levels of severity. For example, the antenna downtilt could be 1°, 0° or even -1°, in contrast to 6° in the normal condition. Besides, the system may experience multiple fault conditions at the same time. Such multi-fault case is more challenging to diagnose and will be addressed in Section III.

B. Network Performance Indicators

The network conditions can be assessed by some basic key performance indicators (KPIs) and performance management counters (PMCs) that are directly related to the user experience. These KPIs and the PMCs are usually measured according to the operations support system (OSS) in each cell. In this paper, we consider downlink transmissions and the following seven KPIs and PMCs.

- Channel quality indicator (CQI): The CQI is an indicator that carries information on the quality of the communication channel. The CQI is proportional to the signal-to-noise ratio (SNR) of the reference signal.
- Rank indicator (RI): The RI is the rank of the channel matrix estimated by the UE. It is an important index for judging correlation of the channel and selection of the transmission layer in downlink data transmission. This information is reported to the eNodeB from the UE. If the UE reports RI as 1 to the eNodeB, the eNodeB switches from the spatial multiplexing mode to the TX diversity mode for downlink data transmission with the UE.
- Received power: The received power is the power level received by each UE, which is given by
$$P_{\mathrm{UE}_i} = P_{\mathrm{BS}_i} G_{\mathrm{BS}_i}^{\mathrm{UE}_i}, \tag{1}$$
where UE_i denotes one of the UEs served by the ith service eNodeB (denoted by BS_i), P_{BS_i} is the transmission power of BS_i, and $G_{\mathrm{BS}_i}^{\mathrm{UE}_i}$ is the channel gain between UE_i and BS_i.
- Signal-to-interference-plus-noise ratio (SINR):The SINR is related to the signal propagation, the interference and the positioning of network transmitters and receivers. Therefore, it is one of the most important indicator to measure the quality of wireless connections. The SINR of each UE is given by
$$\mathrm{SINR}_{\mathrm{UE}_i} = \frac{P_{\mathrm{BS}_i} G_{\mathrm{BS}_i}^{\mathrm{UE}_i}}{\sum_{j=1, j\neq i}^{J} P_{\mathrm{BS}_j} G_{\mathrm{BS}_j}^{\mathrm{UE}_i} + \sigma^2}, \tag{2}$$
where J is the total number of BSs and σ^2 is the additive Gaussian noise power.
- TA counter: The TA counter is defined as the distance between BS and UE. According to [16], 1 TA, which maps the distance between BS and UE, is approximately 78 m.
- Throughput: Throughput is defined as follows:
$$\text{Throughput} = \frac{\text{Transport block size} \cdot \text{RI}}{\text{Transmission time interval}}, \tag{3}$$
where the transport block size is decided by CQI. Every CQI can be mapped to a specific modulation order and coding rate. Besides, 1 transmission time interval (TTI) is equal to 1 millisecond based on the LTE standard.



(a) EU (b) ERP (c) AF (d) CH

Fig. 3. Conditional probability density functions of KPIs given various network fault conditions.

- Link failure indicator: Link failure indicator is dominated by the SINR. In general, when the receive SINR of an UE is less than -5.5 dB, a link failure will occur. Whenever a link failure occurs, the link failure indicator increases by 1. Therefore, the link failure indicator stands for the total number of UEs whose SINR are less than -5.5 dB in a cell.

Whenever a fault occurs, the system KPIs and PMCs would change and deviate from the normal values. In Fig. 3, we present the empirical probability density function (pdf) of several KPIs/PMCs either under the normal condition or under a fault condition. The detailed simulation setting is given in Section IV. From Fig. 3(a), it is clear to see that under the EU condition, the service area of the system is more likely to be larger than under the normality condition. Specifically, as observed from the figure, the mean of the TA counter is higher than that under the normal condition. The ERP condition reduces the transmission power of the base stations. Thus, as shown in Fig. 3(b), the pdf of the received power of UEs is shifted left compared to that under the normal condition. We can also observe from Fig. 3(c) that the AF condition causes the RI to be one because in the simulated worse case there is only one antenna for every UE. The RI is lower than two under the normal condition because the channel correlation would reduce the value of RI. Since the CH condition results in increased attenuation in a few small regions, from Fig. 3(d), one can observe that the CQI is more likely to be small than under the normal condition. From these figures, one can also see how the different severity levels of fault causes may affect the relevant KPIs and PMCs.

As demonstrated in Fig. 3, the seven KPIs and PMCs listed in the previous section can well capture the characteristics for various network conditions. In Section III, we present simple diagnosis algorithms which use the seven KPIs and PMCs as inputs and output the diagnosis result as either the normality condition or one of the fault conditions. The proposed diagnosis algorithms are composed by the classical SVM and SNN. Before presenting the proposed algorithms, let us briefly review the SVM and SNN.

C. SVM and Rank-SVM

In the machine learning literature, SVM is one of most powerful supervised learning models for data classification tasks. Here, let us consider a more general SVM model, called rank-SVM [21], that can handle multi-class and multi-label classification tasks, i.e., one sample may have more than one label and thus belong to more than one class. The rank-SVM generalizes the conventional multi-class SVM [25], and it reduces to the multi-class SVM when each of the samples solely belongs to one class.

Let $\boldsymbol{x}_m \in \mathbb{R}^N,\ m = 1, \ldots, M,$ be the sequence of N-dimensional training samples. Here, each $\boldsymbol{x}_m$ contains N features; for example the seven KPI and PMCs ($N = 7$) listed in the previous subsection for network fault diagnosis. Each of the $\boldsymbol{x}_m$ has a different class label set $\mathcal{Y}_m \subset \{1, \ldots, K\}$, where K is the total number of classes (e.g., fault cases). In particular, $k \in \mathcal{Y}_m$ indicates that sample $\boldsymbol{x}_m$ belongs to class k. In multi-label classification, the size of $\mathcal{Y}_m$, i.e., $|\mathcal{Y}_m|$, can be larger than one. SVM was proposed to maximize the margin between the training patterns and the decision boundary. Specifically, let $\boldsymbol{w}_k \in \mathbb{R}^N$ be the linear classifier for class k, and b_k be a bias coefficient. For each sample $\boldsymbol{x}_m$ with $k \in \mathcal{Y}_m$, the margin can be expressed as [21]

$$\min_{k \in \mathcal{Y}_m, \ell \notin \mathcal{Y}_m} \frac{< \boldsymbol{w}_k - \boldsymbol{w}_\ell, \boldsymbol{x} > + b_k - b_\ell}{\|\boldsymbol{w}_k - \boldsymbol{w}_\ell\|_2}, \tag{4}$$

where $< \cdot, \cdot >$ represents the inner product.

After normalizing $< \boldsymbol{w}_k - \boldsymbol{w}_\ell, \boldsymbol{x}_m > + b_k - b_\ell \geq 1$ for all training data, the margin maximization formulation of SVM is given by

$$\max_{\substack{\boldsymbol{w}_k,\\ k=1,2,\ldots,K}} \min_{\substack{k\in\mathcal{Y}_m,\ell\notin\mathcal{Y}_m\\ m=1,\ldots,M}} \frac{1}{\|\boldsymbol{w}_k - \boldsymbol{w}_\ell\|^2},$$

$$\text{subject to} < \boldsymbol{w}_k - \boldsymbol{w}_\ell, \boldsymbol{x}_m > + b_k - b_\ell \geq 1,$$
$$\forall k \in \mathcal{Y}_m, \ell \notin \mathcal{Y}_m,\ m = 1, 2, \ldots, M, \quad (5)$$

One can employ the existing optimization algorithms for traditional SVM to solve (5) [26]. For a given data sample $\boldsymbol{x}$, the output of the SVM can be expressed as

$$\boldsymbol{s} = \mathrm{SVM}(\boldsymbol{x}) \triangleq \boldsymbol{W}\boldsymbol{x} + \mathbf{b}, \quad (6)$$

where $\boldsymbol{W} = [\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_K]^T,\ \boldsymbol{b} = [b_1, b_2, \ldots, b_K]^T,$ and the superscript T indicates the transpose.

In multi-class (single-labeled) SVM, the sample $\boldsymbol{x}$ is classified into class k if output of the SVM $s_k = \max\{s_1, s_2, \ldots, s_K\}$. In the rank-SVM, a threshold function $t(\boldsymbol{x})$ is also learned, and the sample $\boldsymbol{x}$ is classified into class k if $s_k \geq t(\boldsymbol{x})$, which allows $\boldsymbol{x}$ to be classified into more than one label [21]. As will be shown shortly, we will use $\boldsymbol{s}$ as a feature vector extracted by SVM and combine it with the outcome of SNN for achieving better network diagnosis performance.

D. Softmax Neural Network

The artificial neural network (ANN) [27] is an information processing model inspired by biological neural systems. The feedforward neural network (FNN) is the simplest type of ANN, which is known as the multilayer perceptron (MLP), and consists of an input layer, an output layer, and one hidden layer. In machine learning, FNN can handle both regression and classification problems. For the multi-class classification task, the softmax function is usually used to approximate the posterior probability of data samples, for which we call it SNN. Specifically, according to the Bayesian theorem, the probability of a sample $\boldsymbol{x}_m$ belonging to class k is

$$P(k \in \mathcal{Y}_m|\boldsymbol{x}_m) = \frac{P(\boldsymbol{x}_m|k \in \mathcal{Y}_m)P(k \in \mathcal{Y}_m)}{\sum_{j=1}^{K} P(\boldsymbol{x}_m|j \in \mathcal{Y}_m)P(j \in \mathcal{Y}_m)}$$
$$\approx \frac{\exp(h_k(\boldsymbol{x}_m))}{\sum_{j=1}^{K} \exp(h_j(\boldsymbol{x}_m))}, \quad (7)$$

where $h_1(\boldsymbol{x}), \ldots, h_K(\boldsymbol{x})$ are the parametric nonlinear mapping functions of the SNN. Denote $\boldsymbol{h}$ as the network parameter vector of the SNN.

An approach to determining the network parameters is to maximize the joint posterior probability of the training data samples. In particular, the training of SNN involves solving the following problem:

$$\min_{\boldsymbol{h}} - \sum_{m=1}^{M}\sum_{k=1}^{K} Y_{m,k} \log\left(\frac{\exp(h_k(\boldsymbol{x}_m))}{\sum_{j=1}^{K}\exp(h_j(\boldsymbol{x}_m))}\right) + \frac{\lambda}{2}\|\boldsymbol{h}\|_2^2, \quad (8)$$

where $Y_{m,k} \in \{+1, -1\}$ is equal to 1 if $k \in \mathcal{Y}_m$ and -1 otherwise, and $\frac{\lambda}{2}\|\boldsymbol{h}\|_2^2$ is a regularization factor for avoiding over-fitting.

III. Proposed ML Based Diagnosis Algorithms

In this section, we propose two algorithms to diagnose the network fault conditions. The proposed algorithms rely on a judicious concatenation of the classical SVM and SNN reviewed in the previous section. The first diagnosis algorithm considers the scenario where there is only one fault condition occurring at a time, but the fault may exhibit different levels of severity. The second diagnosis algorithm further deals with the scenario with multiple fault conditions occurring simultaneously, and the training data may not have the complete multi-label information. In Section III-A, we present the data pre-processing. The proposed two diagnosis algorithms are presented in Section III-B and Section III-C, respectively.

A. Data Preprocessing

Since the heterogeneous KPIs and PMCs are used as features for the diagnosis task, to achieve good classification performance, it is essential to process the data before performing any training and diagnosis. The preprocessing stage consists of two steps 1) normalization and 2) weight multiplication.

The normalization step makes all KPIs and PMCs to have the same range of values between 0 and 1 after the normalization. This prevents some KPIs and PMCs that have particularly large range of values from dominating the outcome of learning. Let x_{im} be the ith feature (i.e., KPI and PMC) of the mth training sample, for $i = 1, \ldots, 7$, and $m = 1, \ldots, M$, where M is the size of the training set. Then the data are normalized by

$$\tilde{x}_{im} = \frac{x_{im}}{\max\{x_{i1}, \ldots, x_{iM}\}},\ \forall i, m. \quad (9)$$

While the normalization step normalizes the range of the features, some of the features can be particularly sensitive to some of the fault cases. The importance of these kinds of discriminatory features thus should be emphasized by multiplying them with a proper weight to enhance the diagnosis performance. Specifically, if the statistical property of some feature i can be significantly deviated under certain fault condition, it implies that the sequence of $\tilde{x}_{im}$, $m = 1, \ldots, M$, would contain some "outlier" samples. To identity these key features, we employ the Hampel's identifier method [28] which uses the following identifier for outlier detection:

$$z_i = \frac{\mathrm{MEAN}\left(\left\{\tilde{x}_{im} - \mathrm{MEDIAN}(\{\tilde{x}_{in}\}_{n=1}^{M})\right\}_{m=1}^{M}\right)}{\mathrm{MAD}(\{\tilde{x}_{in}\}_{n=1}^{M})/\mathrm{TH}}. \quad (10)$$

Here, for a sequence of $y_1, \ldots, y_M$, $\mathrm{MEDIAN}(\{y_m\}_{m=1}^{M})$ and $\mathrm{MEAN}(\{y_m\}_{m=1}^{M})$ represent the median value and the mean value of the sequence, respectively, and $\mathrm{MAD}(\{y_m\}_{m=1}^{M})$, which stands for the median absolute deviation, is the median value of the sequence $|y_m - \mathrm{MEDIAN}(\{y_n\}_{n=1}^{M})|$, $m = 1, \ldots, M$. The term TH is a threshold parameter and is set to 0.6745. By [28], KPI/PMC i would be identified to have outliers if $z_i > 3.5$.

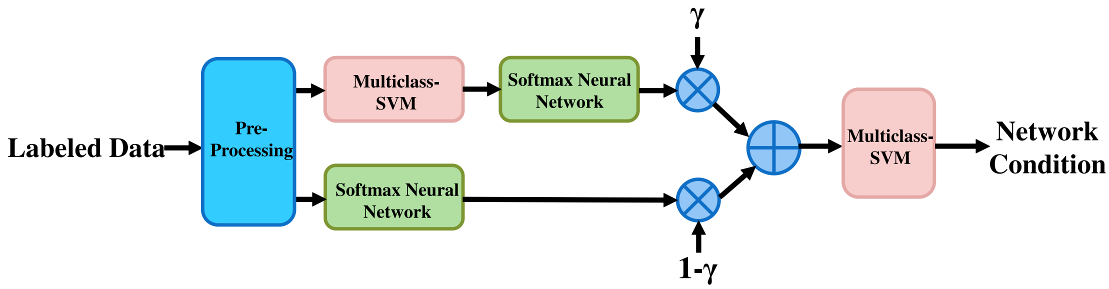


Fig. 4. The structure of proposed Algorithm 1.

Suppose that KPI/PMC i is identified to contain outliers, we multiply the KPI/PMC i in the training sequence $\{\tilde{x}_{im}\}_{m=1}^{M}$ with a weight [20]

$$w_i = c_i^{\beta} r_i, \tag{11}$$

where

$$r_i = \frac{\max_{i=1,\ldots,d}\{\mathrm{MEAN}(\{p_i^{(k)}\}_{k=1}^{K})\}}{\mathrm{MEAN}(\{p_i^{(k)}\}_{k=1}^{K})}, \tag{12}$$

$$c_i = \sqrt{\sum_{k=1}^{K}\left(r_i p_i^{(k)} - r_i \mathrm{MEAN}(\{p_i^{(k)}\}_{k=1}^{K})\right)^2}, \tag{13}$$

and $p_i^{(k)} = \mathrm{MEAN}(\{\tilde{x}_{im}^{(k)}\}_m)$, in which $\tilde{x}_{im}^{(k)}$ denotes the ith feature of the training sample that is labeled with fault condition k. As seen in (12), the constant r_i is used to balance the contribution of each feature since the weighted feature sequence $\{r_i\tilde{x}_{im}\}_{m=1}^{M}, i = 1,\ldots,7$, would have the same mean. The constant c_i in (13) represents the standard deviation of $\{r_i p_i^{(k)}\}_{k=1}^{K}$. The parameter β ranges from 1.1 to 1.5 by rule of thumb [29]. Multiplying the feature sequence $\{r_i\tilde{x}_{im}\}_{m=1}^{M}$ with c_i^{β} thereby emphasizes the importance of this feature for distinguishing between different network conditions. As a result, the weighted feature sequence $\hat{x}_{im} = w_i\tilde{x}_{im}, m = 1,\ldots,M$, would provide better separability for network fault diagnosis, which will be verified in Section IV.

B. Proposed Diagnosis Algorithm

We assume that there is only one fault condition occurring at a time, but the fault may exhibit different levels of severity. We aim to develop a diagnosis system that is able to correctly diagnose the fault cause not only for the severe case but also for the mild case (see Fig. 3). To achieve this, we propose to fully utilize the distinct but complementary natures of SNN and SVM. Specifically, the SNN is arguably more powerful than SVM, provided that the feature size is large and the training set is large. However, since the considered network diagnosis problem has a small feature size, the SNN may not always perform better. Let us take Fig. 3(b) as an example. From Fig. 3(b), it could be an easy task for both the SVM and SNN to distinguish between the normal condition ($P_{\max} = 46$ dBm) and the worst ERP condition ($P_{\max} = 7$ dBm). However, it is more challenging to distinguish between the normal condition and the milder ERP condition ($P_{\max} = 23$ dBm) since their statistical distribution of received signal powers is close to each other. In the latter case, the data for the two conditions are not fully separated. Since the SVM is a large-margin method which maximizes the margin between one class and the others, it is more robust for non-well-separated data [30]. Thus, it is likely to achieve a better diagnosis performance if both features of SVM and SNN can be fully explored.

To this end, we propose the diagnosis system as displayed in Fig. 4, which consists of two paths after the preprocessing stage. In the lower path, the SNN is used, which outputs a probability vector indicating how likely the data is to belong to each of the fault cases or the normal condition. Let us denote $\boldsymbol{z}_1 = \mathrm{SNN}_1(\hat{\boldsymbol{x}})$ as the probability vector, where $\hat{\boldsymbol{x}}$ is the preprocessed data. While the traditional strategies make the diagnosis decision solely based on the probability vector $\boldsymbol{z}_1$, we treat $\boldsymbol{z}_1$ as a learned feature vector, and will combine it with the other feature vector learned by SVM in the upper path of Fig. 4.

Let $\boldsymbol{s}_1 = \mathrm{SVM}_1(\hat{\boldsymbol{x}})$ be the output of multi-class SVM in the upper path. Traditionally the index of the largest element in $\boldsymbol{s}_1$ indicates which fault case is present. Again, we treat $\boldsymbol{s}_1$ as a learned feature vector. To obtain a probability vector that has the same meaning as $\boldsymbol{z}_1$, we pass $\boldsymbol{s}_1$ to another SNN in the upper path, which yields the probability vector $\boldsymbol{z}_2 = \mathrm{SNN}_2(\boldsymbol{s}_1)$.

As mentioned, the SNN is more powerful if the feature size is large. Based on this, we linearly combine $\boldsymbol{z}_1$ in the lower path and $\boldsymbol{z}_2$ in the upper path by multiplying them with γ and $1-\gamma$, respectively:

$$\boldsymbol{z}_3 = \gamma\boldsymbol{z}_2 + (1-\gamma)\boldsymbol{z}_1. \tag{14}$$

Here γ is set to the inverse of the feature size (i.e., $\gamma = 1/7$), and therefore the SNN would play a more important role if the feature size is larger. The combined feature vector $\boldsymbol{z}_3$ then is fed into the SVM for fault diagnosis.

The training and testing procedures of the proposed diagnosis algorithm are summarized in Algorithm 1 below. Specifically, each of the SNNs and multi-class SVMs in Fig. 4 are sequentially trained. The numerical testing results are presented in Section IV.

C. Proposed Multi-Fault Diagnosis Algorithm

In the previous subsection, we have assumed that there is only one fault condition occurring at the same time. In practice, there could exist multiple faults simultaneously, and therefore there is a need for the diagnosis system to be able to diagnose

Algorithm 1: Proposed Diagnosis Algorithm.

Training:

A1. **Input:** Given the training set $\{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_M, y_M)\} \subset (\mathcal{X} \times \mathcal{Y})^M$ where each $y_m \in \mathcal{Y}$ is the label of the input feature (KPI/PMC) vector $\boldsymbol{x}_m$, for all $m = 1, \ldots, M$.

A2. **Normalization:** Normalized the data set by (9) and obtain $\{(\tilde{\boldsymbol{x}}_1, y_1), \ldots, (\tilde{\boldsymbol{x}}_M, y_M)\}$.

A3. **Weight calculation:** Perform outlier detection by (10) and calculate the weights by (11)-(13).

A4. **Weight multiplication:** Multiply the training data with the weights; obtain $\{(\hat{\boldsymbol{x}}_1, y_1), \ldots, (\hat{\boldsymbol{x}}_M, y_M)\}$

A5. **Feature learning:** Firstly, train the SNN (e.g., using function `patternnet, train` in `MATLAB`) in the lower path in Fig. 4 with $\{(\hat{\boldsymbol{x}}_1, y_1), \ldots, (\hat{\boldsymbol{x}}_M, y_M)\}$. The output of the trained SNN for $\{(\hat{\boldsymbol{x}}_1, y_1), \ldots, (\hat{\boldsymbol{x}}_M, y_M)\}$ are denoted as $\{(\boldsymbol{z}_{1,1}, y_1), \ldots, (\boldsymbol{z}_{M,1}, y_M)\}$, where $\boldsymbol{z}_{m,1} = \mathrm{SNN}_1(\hat{\boldsymbol{x}}_m)$, $m = 1, \ldots, M$. Secondly, train the multi-class SVM (e.g., using function `fitcecoc` in `MATLAB`) in the upper path in Fig. 4 using $\{(\hat{\boldsymbol{x}}_1, y_1), \ldots, (\hat{\boldsymbol{x}}_M, y_M)\}$, and obtain the output of the trained SVM as $\{(\boldsymbol{s}_{1,1}, y_1), \ldots, (\boldsymbol{s}_{M,1}, y_M)\}$, where $\boldsymbol{s}_{m,1} = \mathrm{SVM}_1(\hat{\boldsymbol{x}}_m)$, $m = 1, \ldots, M$. Thirdly, train the SNN in the upper path in Fig. 4 with $\{(\boldsymbol{s}_{1,1}, y_1), \ldots, (\boldsymbol{s}_{M,1}, y_M)\}$ and obtain the output $\{(\boldsymbol{z}_{1,2}, y_1), \ldots, (\boldsymbol{z}_{M,2}, y_M)\}$, where $\boldsymbol{z}_{m,2} = \mathrm{SNN}_2(\boldsymbol{s}_{m,1})$, $m = 1, \ldots, M$. Lastly, obtain the combined feature vectors $\boldsymbol{z}_{m,3} = \gamma \boldsymbol{z}_{m,2} + (1-\gamma)\boldsymbol{z}_{m,3}$, $m = 1, \ldots, M$.

A6. **Classifier training:** Train the SVM at the output stage using $\{(\boldsymbol{z}_{1,3}, y_1), \ldots, (\boldsymbol{z}_{M,3}, y_M)\}$.

Testing:

B1. **Input:** Given the testing set $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots\}$.

B2. **Normalization and weight multiplication:** Perform data normalization and weight multiplication to obtain $\{\hat{\boldsymbol{x}}_1, \hat{\boldsymbol{x}}_2, \ldots\}$.

B3. **Feature extraction:** Obtain the feature-extracted data as

$$\boldsymbol{z}_{n,3} = \gamma \mathrm{SNN}_2(\mathrm{SVM}_1(\hat{\boldsymbol{x}}_n)) + (1-\gamma)\mathrm{SNN}_1(\hat{\boldsymbol{x}}_n),$$

for $n = 1, 2 \ldots$.

B4. **Fault diagnosis:** Diagnose the fault condition by the multi-class SVM using $\boldsymbol{z}_{n,3}$ as input, for $n = 1, 2 \ldots$.

multiple faults if they exist. This problem then belongs to the so called multi-label classification problem [25]. In addition, the labeling of training samples may be incomplete since in practice the engineers may only label one of the severest faults even when there are actually multiple faults present at the same time. That is, the training data set could have partially missing labels.

The proposed algorithm for the multiple fault diagnosis problem with missing labels is shown in Fig. 5. Firstly, to achieve the multi-label diagnosis, we replace the multi-class SVMs in Fig. 4 by the rank-SVM [21], which is a variant of SVM and is able to perform multi-label classification. Secondly, to overcome the missing label problem, we incorporate a re-training step in the training stage. Specifically, after finishing training the SNNs and rank-SVMs in Fig. 5, according to the diagnosis outcome of rank-SVM at the output stage for training data, we add to the data the label of the newly diagnosed fault that is not labeled in the original training set. Then, the re-labeled training set is used to train the diagnosis again. Numerical results in Section IV will show that the retraining step can significantly improve the multi-fault diagnosis performance. We summarize the training steps of the proposed multi-fault diagnosis algorithm below.

Algorithm 2: Proposed Multi-Fault Diagnosis Algorithm.

Training:

C1. **Input:** Given the training set $\{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_M, y_M)\} \subset (\mathcal{X} \times \mathcal{Y})^M$ where each $y_m \in \mathcal{Y}$ is the label of the input feature (KPI/PMC) vector $\boldsymbol{x}_m$, for all $m = 1, \ldots, M$.

C2. **Preprocessing and feature extraction:** Perform the training steps A2 to A5 in Algorithm 1, except replacing the multi-class SVM by rank-SVM; obtain the combined feature vectors $\boldsymbol{z}_{m,3} = \gamma \boldsymbol{z}_{m,2} + (1-\gamma)\boldsymbol{z}_{m,3}$, $m = 1, \ldots, M$.

C3. **Classifier training:** Train the rank-SVM at the output stage using $\{(\boldsymbol{z}_{1,3}, y_1), \ldots, (\boldsymbol{z}_{M,3}, y_M)\}$.

C4. **Diagnosis and re-labeling:** Perform multi-fault diagnosis using $\{\boldsymbol{z}_{1,3}, \ldots, \boldsymbol{z}_{M,3}\}$ as the input of the rank-SVM at the output stage in Fig. 5, and based on the outcome, re-label the data. Denote the re-labeled training data set as $\{(\boldsymbol{x}_1, \hat{\boldsymbol{y}}_1), \ldots, (\boldsymbol{x}_M, \hat{\boldsymbol{y}}_M)\}$.

C5. **Re-training:** Perform C2 and C3 using the re-labeled training set $\{(\boldsymbol{x}_1, \hat{\boldsymbol{y}}_1), \ldots, (\boldsymbol{x}_M, \hat{\boldsymbol{y}}_M)\}$.

Testing:

D1. **Input:** Given the testing set $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots\}$.

D2. **Normalization and weight multiplication:** Perform data normalization and weight multiplication to obtain $\{\hat{\boldsymbol{x}}_1, \hat{\boldsymbol{x}}_2, \ldots\}$.

D3. **Feature extraction:** Obtain the feature-extracted data as

$$\boldsymbol{z}_{n,3} = \gamma \mathrm{SNN}_2(\mathrm{SVM}_1(\hat{\boldsymbol{x}}_n)) + (1-\gamma)\mathrm{SNN}_1(\hat{\boldsymbol{x}}_n),$$

for $n = 1, 2 \ldots$.

D4. **Fault diagnosis:** Diagnose the fault condition by the rank-SVM using $\boldsymbol{z}_{n,3}$ as input, for $n = 1, 2 \ldots$.

IV. Simulation Results

In this section, we examine the performance of the proposed network diagnosis algorithms. The LTE-A system level simulation is conducted based on an open-source Matlab simulator created by Vienna University of Technology [31]. The neural network toolbox and statistics and machine learning toolbox of Matlab are used to implement the proposed Algorithm 1 and Algorithm 2.

The system setting and channel model are described as follows.

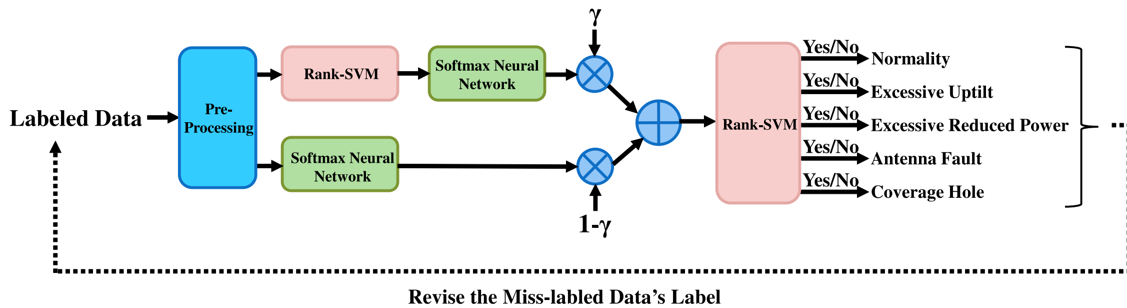


Fig. 5. The structure of proposed Algorithm 2.

TABLE I
PARAMETERS CONFIGURATION

Parameters	Configuration
Small scale channel model	Typical urban
Macro inter-site distance	500m
Number of transmit antennas N_t at eNodeBs	4
Number of receive antennas N_r at UEs	2
Height of eNodeBs	2.5m
Height of UEs	1.5m
Number of eNodeBs	7
Number of UEs per cell	20
Speed of UEs	1.5m/s
Channel bandwidth W	20MHz
Downlink carrier frequency f_c	2.14GHz
Subframe duration T_f (1TTI)	1ms
Number of resource blocks	100
Type of transmit antenna	Kathrein
Transmission mode	All the Same

TABLE II
SINGLE-FAULT CONDITIONS

Fault Case	Configuration
Normality	$Downtilt = 6°,\ P_{\text{BS}} = 46$ dBm $RI = 2,\ Shadow\ gain = 0$ dB
EU	$Downtilt = [-1, 0.5, 1]°,\ P_{\text{BS}} = 46$ dBm $RI = 2,\ Shadow\ gain = 0$ dB
ERP	$Downtilt = 6°,\ P_{\text{BS}} = [7, 15, 23]$ dBm $RI = 2,\ Shadow\ gain = 0$ dB
AF	$Downtilt = 6°,\ P_{\text{BS}} = 46$ dBm $RI = 1,\ Shadow\ gain = 0$ dB
CH	$Downtilt = 6°,\ P_{\text{BS}} = 46$ dBm $RI = 2,\ Shadow\ gain = [-20, -16.9, -13.9]$ dB

A. System Configuration and Channel Model

In our simulation, the parameter configuration of the simulated network follows Table I.

The macrocell inter-site distance is 500 m. The speed of each UE is 1.5 m/s. The number of UEs is around 20 in each cell, and it can change from 1 TTI to another. Since the UEs can move randomly, handover between macrocells is considered. The system operates in the FDD mode, and synchronization is assumed perfect. Each BS transmits signals to the served UEs, and each UE feeds back KPIs to the serving BS. The average heights of eNodeBs and UEs are about 20 m and 1.5 m, respectively. Downlink frequency is 2.14 GHz, as specified by 3GPP [32]. The bandwidth of each UE is 20 MHz and the total number of resource blocks is 100. Thus each UE at most occupies five resource blocks. The number of antenna ports of each eNodeB and UE are four and two, respectively. The transmission modes are as defined by 3GPP [24].

According to [33], the path loss model is given by

$$L = 40(1 - 4 \cdot 10^{-3} \cdot \text{AH}) \cdot \log_{10}(R) - 18 \log_{10}(\text{AH}) + 21 \log_{10}(f) + 80 \text{ dB}, \quad (15)$$

where AH is the antenna height of eNodeB in meters (m), R is the distance between eNodeB and UE in kilometers (km), and f is the carrier frequency in MHz. Shadow fading is also considered, and the spatial correlations of shadow fading follows the model in [34]. The normalized correlation function can be written as follows:

$$r(x) = e^{-\alpha x},\ x > 0, \quad (16)$$

where $e^{-\alpha x}$ is the correlation coefficient between two points spaced by $x = 1$. In [35], a value of $\alpha = 1/20$ is suggested for typical urban cellular.

B. Synthetic Data Generation

In the single-fault case, we respectively consider scenarios for each one of the five fault causes described in Section II-A. The detailed parameter settings are shown in Table II. Note that for scenarios with EU, ERP and CH, we consider three different levels of severity (severe, medium and slight) for the antenna tilt, BS transmission power and shadowing gain, respectively. During the training, we consider only the severe case, that is, we choose downtilt as $-1°$ for the EU condition, $P_{BS} = 7$ dBm for the ERP condition, shadowing gain as -20 dB for the CH condition, and that all UEs have one broken antenna for the AF condition. However, for testing, we not only consider the data with the most severe fault conditions (Table VII) but also test data with all kinds of levels of severity (Table VIII and Table IX).

TABLE III
MULTI-FAULT CONDITIONS

Fault Case	Configuration
EU+CH	$Downtilt = -1°,\ P_{BS} = 46$ dBm $RI = 2,\ Shadow\ gain = -20$ dB
EU+AF	$Downtilt = -1°,\ P_{BS} = 46$ dBm $RI = 1,\ Shadow\ gain = 0$ dB
EU+ERP	$Downtilt = -1°,\ P_{BS} = 7$ dBm $RI = 2,\ Shadow\ gain = 0$ dB
AF+CH	$Downtilt = 6°,\ P_{BS} = 46$ dBm $RI = 1,\ Shadow\ gain = -16$ dB

TABLE IV
THE TRAINING PARAMETER OF SNN

Parameters	Configuration
Loss function	Cross entropy
Learning rate	1e-6
Batch size	25
Maximum number of epoch	50
Early stopping	10-fold cross-validation
Number of hidden layer	1
Number of input neurons	Data feature size
Number of hidden neurons	Data feature size
Number of output neurons	Data class size
Transfer function of hidden neurons	Linear
Transfer function of input neurons	Linear
Transfer function of output neurons	Softmax

In the multi-fault case, we consider scenarios with four different combinations of fault causes, as shown in Table III. There are two experiments. In the first experiment, we use the same training set as the single-fault case to train Algorithm 2 (without retraining), but use multi-fault data in the testing stage (Table X). In the second experiment, we blend 5% single-labeled multi-fault data in the training set to train Algorithm 2 (with retraining), and evaluate the diagnosis performance using multi-fault data during testing (Table XI).

In the simulations, under each scenario, we repeat the system level simulation of LTE-A for 150 trials. In each trial, the UEs receive signals from the BS for 5 TTIs and feed back the measured 750 KPI/PMC vectors to the BS. The training data are obtained by averaging for every 10 KPI/PMC vectors, and thus there are 75 training data samples of each fault cases. The testing data are obtained by 50 trials of simulations, and thereby there are 25 testing data samples of each fault cases. The functions `patternnrt`, `train` and `fitcecoc` in `Matlab` are used to implement SNN, multi-class SVM and rank-SVM. Their detailed settings in the simulations are listed in Table IVto Table VI.

C. Performance Results: Single-Fault Case

In addition to the proposed algorithms, we also implemented the network diagnosis algorithm proposed in [11]. The diagnosis

TABLE V
THE TRAINING PARAMETER OF MULTI-CLASS SVM

Parameters	Configuration
Kernal function type	Linear
Number of Class	Data class size
Multiclass method	One-vs-all

TABLE VI
THE TRAINING PARAMETER OF RANK-SVM

Parameters	Configuration
Kernal function type	Linear
Number of Class	Data class size

TABLE VII
SINGLE-FAULT CONDITIONS

(a) Proposed Algorithm 1

Fault Cause	Normal	EU	ERP	AF	CH
Normal	**90.1%**	0%	7.6%	0%	0%
EU	0%	**100%**	0%	0%	0%
ERP	9.9%	0%	**92.4%**	0%	0%
AF	0%	0%	0%	**100%**	0%
CH	0%	0%	0%	0%	**100%**

(b) Diagnosis algorithm in [11]

Fault Cause	Normal	EU	ERP	AF	CH
Normal	**48.9%**	0%	**62.3%**	24.3%	0%
EU	44.6%	**97.3%**	2.3%	37.2%	7.2%
ERP	0%	2.7%	35.4%	0%	0%
AF	1.6%	0%	0%	**38.5%**	0%
CH	4.9%	0%	0%	0%	**92.8%**

(c) Proposed Algorithm 1 without weight multiplication

Fault Cause	Normal	EU	ERP	AF	CH
Normal	**78.9%**	0%	14.3%	5.8%	1.6%
EU	0%	**98.5%**	0%	0%	0.6%
ERP	13.9%	0%	**82.6%**	2.1%	0.9%
AF	6.4%	0.8%	3.1%	**92.1%**	0.5%
CH	0.8%	0.7%	0%	0%	**96.4%**

(d) Proposed Algorithm 1 with $\gamma = 1$

Fault Cause	Normal	EU	ERP	AF	CH
Normal	**85.1%**	0%	13.8%	9.9%	0%
EU	0%	**99.7%**	0%	0%	3.6%
ERP	14.9%	0%	**86.2%**	0%	0%
AF	10%	0%	0%	**90.1%**	0%
CH	0%	0.3%	0%	0%	**96.4%**

(e) Proposed Algorithm 1 with $\gamma = 0$

Fault Cause	Normal	EU	ERP	AF	CH
Normal	**86.4%**	0%	11.2%	0%	0%
EU	0%	**99.9%**	0%	0%	0%
ERP	13.6%	0%	**88.8%**	0%	0%
AF	0%	0%	0%	**100%**	0%
CH	0%	0.1%	0%	0%	**100%**

TABLE VIII
PERFORMANCE UNDER DIFFERENT LEVELS OF EU SEVERITY

(a) Proposed Algorithm 1

Fault Cause	EU(severe)	EU(medium)	EU(slight)
Normal	**0%**	**2.3%**	**21.4%**
EU	**100%**	**95.5%**	**71.8%**
ERP	0%	2.1%	6.7%
AF	0%	0%	0%
CH	0%	0.1%	0.1%

(b) Diagnosis algorithm in [11]

Fault Cause	EU(severe)	EU(medium)	EU(slight)
Normal	0%	0%	11.8%
EU	**97.3%**	**93.2%**	**65.3%**
ERP	**2.7%**	2.9%	10.3%
AF	0%	0%	0%
CH	0%	**3.9%**	**12.6%**

(c) Proposed Algorithm 1 with $\gamma = 1$

Fault Cause	EU(severe)	EU(medium)	EU(slight)
Normal	0%	**3.1%**	**27.1%**
EU	**99.7%**	**94.7%**	**65.2%**
ERP	0%	2.1%	7.6%
AF	0%	0%	0%
CH	**0.3%**	0.1%	0.1%

(d) Proposed Algorithm 1 with $\gamma = 0$

Fault Cause	EU(severe)	EU(medium)	EU(slight)
Normal	0%	2.9%	15.6%
EU	**99.9%**	**90.5%**	**57.1%**
ERP	0%	**5.7%**	**26.2%**
AF	0%	0%	0%
CH	**0.1%**	0.9%	1.1%

TABLE IX
PERFORMANCE UNDER DIFFERENT LEVELS OF ERP SEVERITY

(a) Proposed Algorithm 1

Fault Cause	ERP(severe)	ERP(medium)	ERP(slight)
Normal	**7.6%**	**8.6%**	**9.1%**
EU	0%	0%	0%
ERP	**92.4%**	**91.4%**	**90.9%**
AF	0%	0%	0%
CH	0%	0%	0%

(b) Diagnosis algorithm in [11]

Fault Cause	ERP(severe)	ERP(medium)	ERP(slight)
Normal	**62.3%**	**52.1%**	**60.2%**
EU	2.3%	4.2%	2.1%
ERP	**35.4%**	**43.5%**	**36.2%**
AF	0%	0%	0%
CH	0%	0.2%	1.5%

(c) Proposed Algorithm 1 with $\gamma = 1$

Fault Cause	ERP(severe)	ERP(medium)	ERP(slight)
Normal	**13.8%**	**14.1%**	**15.1%**
EU	0%	0%	0%
ERP	**86.2%**	**85.9%**	**84.9%**
AF	0%	0%	0%
CH	0%	0%	0%

(d) Proposed Algorithm 1 with $\gamma = 0$

Fault Cause	ERP(severe)	ERP(medium)	ERP(slight)
Normal	**11.2%**	**15.6%**	**20.1%**
EU	0%	0%	0%
ERP	**88.8%**	**84.4%**	**79.9%**
AF	0%	0%	0%
CH	0%	0%	0%

algorithm in [11] relies on the expert knowledge to learn the normal behavior of the network and the impacts of different fault causes on different KPIs and PMCs. Whenever a KPI or PMC significantly deviates from the normal behavior, the algorithm gives scores to relevant fault causes and make a diagnosis decision.

The performance comparison results are displayed in Table VII. The testing data contain uniform samples with severe single-fault conditions. The diagonal entries in the table are the diagnosis accuracy under different network conditions, while the off-diagonal entries are the probability that the diagnosis algorithm mistakes one network condition for another. For example, in Table VII(a), the proposed Algorithm 1 has a probability of 9.9% to wrongly diagnose the normal condition as ERP, and has a probability of 7.6% to wrongly diagnose ERP as the normal condition. Nevertheless, the proposed Algorithm 1 can almost surely and correctly detect EU, AF and CH. By contrast, one can see from Table VII(b) that the diagnosis algorithm in [11] does not perform well, especially under ERP, AF and the normal condition. The algorithm in [11] has a high probability of 62.3% to wrongly diagnose ERP as the normal condition.

In Table VII(c), we present the performance of the proposed Algorithm 1 without the weight multiplication in (11). By comparing with Table VII(a), one can observe that the performance degrades. In particularly, it has a higher probability for the proposed algorithm to wrongly diagnose one fault condition as another. This demonstrates that the weight multiplication in (11) is indeed helpful in improving the diagnosis performance. In Table VII(d) and Table VII(e), we respectively present the performance of the proposed Algorithm 1 with $\gamma = 1$ and $\gamma = 0$. As seen, both have reduced diagnosis performance. However, the proposed algorithm with $\gamma = 0$ (SNN) is better than that with $\gamma = 1$ (SVM).

In Table VIII, we present the comparison results where the testing data contain samples with three different levels of severity of EU condition. As seen, for the severe EU case, the proposed Algorithm 1 achieves 100% accuracy and the method in [11] has 97.3% accuracy. However, for the medium case and the

TABLE X
MULTI-FAULT CONDITIONS

(a) Proposed Algorithm 2 without retraining

Fault Cause	EU+CH	EU+AF	EU+ERP	AF+CH
Normal	0%	0%	14.7%	11.5%
EU	**90.5%**	**100%**	**100%**	13.3%
ERP	9.5%	27.3%	**67.4%**	0%
AF	0%	**70.5%**	14.2%	**75.2%**
CH	**100%**	2.2%	3.7%	**100%**
Accuracy	90.5%	70.5%	66.1%	73.5%

(b) Diagnosis algorithm in [11]

Fault Cause	EU+CH	EU+AF	EU+ERP	AF+CH
Normal	12.7%	34.6%	0%	0%
EU	**90.1%**	**100%**	**100%**	**100%**
ERP	0%	0%	0%	0%
AF	0%	**62.2%**	49.3%	0%
CH	**97.2%**	3.2%	**50.7%**	**100%**
Accuracy	87.9%	59.4%	0%	0%

TABLE XI
MULTI-FAULT CONDITIONS WITH 5% MISLABELED ERRORS

(a) Proposed Algorithm 2 without retraining

Fault Cause	EU+CH	EU+AF	EU+ERP	AF+CH
Accuracy	90.5%	70.5%	65.4%	71.3%

(b) Proposed Algorithm 2

Fault Cause	EU+CH	EU+AF	EU+ERP	AF+CH
Accuracy	92.5%	83.7%	72.2%	84%

slight case, the proposed Algorithm 1 exhibits better robustness than the method in [11]. Interestingly, from Table VIII(c) and Table VIII(d), one can see that the proposed algorithm with $\gamma = 1$ (SVM) outperforms the proposed algorithm with $\gamma = 0$ (SNN). This verifies that the SVM (which is a large margin method) can have a better classification capability if the data are not well separated, as well as justifying the effectiveness of our proposed Algorithm 1 in Fig. 4 that combines features extracted by SVM and SNN. In Table IX, the comparison results obtained by testing samples with three different levels of severity of ERP condition are presented, and similar observations can be made.

D. Performance Results: Multiple-Fault Case

Let us consider the multi-fault conditions in Table III. We use the same training set as the single-fault case to train Algorithm 2 (without retraining), but use multi-fault data in the testing stage. Table X presents the simulation results of the proposed Algorithm 2 without retraining as well as the diagnosis algorithm in [11]. Both the overall diagnosis accuracy and the probability that each single fault is diagnosed are shown. As seen, while the training data have only single labels, the proposed Algorithm 2 using rank-SVM can still perform multi-label classification, and yield much higher diagnosis accuracy than the diagnosis algorithm in [11]. In Table X(b), one can see that the diagnosis algorithm in [11] can hardly correctly find the two-fault conditions of EU + ERP and AF + CH. In fact, for the EU + ERP condition, the algorithm in [11] can always detect the presence of the EU condition but almost cannot identify the ERP condition. Similarly, for the AF + CH condition, the algorithm in [11] always diagnose the fault as EU + CH in the tested experiments.

To demonstrate the efficacy of the retraining process, we consider the case that 5% of the training data have multiple faults but are (randomly) labeled with one fault only. The training data is used to train the proposed Algorithm 2 with and without the retraining step, respectively. The simulation results are presented in Table XI. By comparing Table XI(a) with Table X(a), one can see that the presence of mislabeled training data can degrade the performance of the proposed Algorithm 2 (without retraining) slightly. Intriguingly, as observed from Table XI(b), the retraining can greatly improve the diagnosis performance, especially for the cases of EU + AF, EU + ERP and AF + CH.

V. CONCLUSION

In this paper, we have proposed two new diagnosis algorithms for the self-healing system based on machine learning techniques. The proposed algorithms are built by judiciously concatenating the classical SVM and SNN and are able to diagnose the normal condition and fault causes in the wireless network. A preprocessing procedure including data normalization and outlier-based weight multiplication is also proposed. The proposed algorithms not only can diagnose single-fault cause (Algorithm 1) but also multi-fault cause (Algorithm 2) with missing labels. The presented simulation results based on the LTE-A system have shown that the proposed Algorithm 1 can achieve much better diagnosis performance than the existing method in [11]. In particular, the proposed Algorithm 1 is more robust against different levels of severity of fault causes. Moreover, the proposed Algorithm 2 with re-training can further improve the diagnosis performance when there are missing labels in the training set. In the future, it is worthwhile to investigate the mislabeled problem by considering advanced semi-supervised models in the ML literature [36], [37].

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